

readr, tibble va tidyr yordamida Ma'lumotlar Importi

bo'yicha qo'llanma



Rning **tidyverse** **tibble**da saqlanuvchi **toza ma'lumotlar** asosida qurilgan.



Mazkur qo'llanmaning old qismi **readr** yordamida Rda matnli fayllarni o'qishni ko'rsatadi.



Orqa qismi esa, **tibble** yordamida tibblelarni yaratish va **tidyr** yordamida tozalashni ifodalaydi.

Ma'lumotlarning boshqa turlari

Quyidagi paketlar boshqa turdagi fayllarni import qilish uchun xizmat qiladi

- **haven** - SPSS, Stata va SAS fayl
- **readxl** - excel fayllari (.xls va .xlsx)
- **DBI** - ma'lumotlar ombori
- **jsonlite** - json
- **xml2** - XML
- **httr** - Web API
- **rvest** - HTML (Web Scraping)

Yozish funksiyalari

x R obyektini **path** nomli katalogga yozish:

write_csv(x, path, na = "NA", append = FALSE, col_names = !append)

Tibble/df ni vergul bilan ajratilgan faylga

write_delim(x, path, delim = " ", na = "NA", append = FALSE, col_names = !append)

Tibble/df ni ixtiyoriy belgi bilan ajratilgan faylga

write_excel_csv(x, path, na = "NA", append = FALSE, col_names = !append)

Tibble/df ni excel uchun CSV faylga o'tkazish

write_file(x, path, append = FALSE)

Qatorni faylga o'g'irish.

write_lines(x, path, na = "NA", append = FALSE)

Qatorli vektorni faylga o'g'irish.

write_rds(x, path, compress = c("none", "gz", "bz2", "xz"), ...)

Obyektini RDS faylga o'tkazish.

write_tsv(x, path, na = "NA", append = FALSE, col_names = !append)

Tibble/df ni tab bilan ajratilgan faylga o'tkazish.

O'qish funksiyalari

Jadval shaklidagi ma'lumotlarni tibblega aylantirish

Ushbu funksiyalar quyidagi umumiy argumentlarga ega:

read_*(file, col_names = TRUE, col_types = NULL, locale = default_locale(), na = c("", "NA"), quoted_na = TRUE, comment = "", trim_ws = TRUE, skip = 0, n_max = Inf, guess_max = min(1000, n_max), progress = interactive())

a,b,c
1,2,3
4,5,NA

A	B	C
1	2	3
4	5	NA

read_csv()

Vergul bilan ajratilgan fayllar.

read_csv("file.csv")

a;b;c
1;2;3
4;5;NA

A	B	C
1	2	3
4	5	NA

read_csv2()

Nuqta-vergul yordamida ajratilgan fayllar.

read_csv2("file2.csv")

a|b|c
1|2|3
4|5|NA

A	B	C
1	2	3
4	5	NA

read_delim(delim, quote = "\"", escape_backslash = FALSE, escape_double = TRUE)

Ixtiyoriy belgi bilan ajratilgan fayllar.

read_delim("file.txt", delim = "|")

a b c
1 2 3
4 5 NA

A	B	C
1	2	3
4	5	NA

read_fwf(col_positions)

O'zgarmas kenglikli fayllarni o'qish.

read_fwf("file.fwf", col_positions = c(1, 3, 5))

read_tsv()

Tabulyatsiya bilan ajratilgan fayllarni o'qish. Shuningdek,

read_table(). **read_tsv("file.tsv")**

Foydali argumentlar

a,b,c
1,2,3
4,5,NA

Namunaviy fayl

write_csv(path = "file.csv",
x = read_csv("a,b,c\n1,2,3\n4,5,NA"))

A	B	C
1	2	3
4	5	NA

Sarlavhasiz

read_csv("file.csv",
col_names = FALSE)

x	y	z
A	B	C
1	2	3
4	5	NA

Sarlavhali

read_csv("file.csv",
col_names = c("x", "y", "z"))

1	2	3
4	5	NA

Qatorlarni tashlash

read_csv("file.csv",
skip = 1)

A	B	C
1	2	3

Kichik to'plamga o'qish

read_csv("file.csv",
n_max = 1)

A	B	C
1	2	3
NA	NA	NA

Yo'q qiymatlar

read_csv("file.csv",
na = c("4", "5", "."))

Jadval shaklida bo'lmagan ma'lumotlarni o'qish

read_file(file, locale = default_locale())

Faylni yagona Stringga o'qish

read_file_raw(file)

Fayli vektor shaklida o'qish

read_lines(file, skip = 0, n_max = -1L, locale = default_locale(), na = character(), progress = interactive())

Har bir qatorni alohida string sifatida o'qish.

read_lines_raw(file, skip = 0, n_max = -1L, progress = interactive())

Har bir qatorni alohida vektor sifatida o'qish

read_log(file, col_names = FALSE, col_types = NULL, skip = 0, n_max = -1, progress = interactive())

Apache stilidagi log fayllar

Ma'lumot parsingi

readr funksiyalari har bir ustunning turini aniqlashga va mos kelsa o'zgartirishga harakat qiladi (biroq u stringlarni faktorlarga avtomatik tarzda almashtirmaydi).

Quyida keltirilgan xabar, natijadagi har bir ustunning turini quyidagicha ifodalaydi.

```
## Parsed with column specification:
## cols(
##   age = col_integer(),
##   sex = col_character(),
##   earn = col_double()
## )
```

age bu butun son
sex bu belgi
earn bu double toifali son

1. Muammoni aniqlash **problems()**

x <- read_csv("file.csv"); problems(x)

2. Parsingga col_ funksiya bilan ko'rsatma berish

- **col_guess()** - odatiy holda
- **col_character()**
- **col_double()**
- **col_euro_double()**
- **col_datetime**(format = ""). Shuningdek, **col_date**(format = "") va **col_time**(format = "")
- **col_factor**(levels, ordered = FALSE)
- **col_integer()**
- **col_logical()**
- **col_number()**
- **col_numeric()**
- **col_skip()**

**x <- read_csv("file.csv", col_types = cols(
A = col_double(),
B = col_logical(),
C = col_factor()
))**

3. Yoki, belgi sifatida o'qib, parse_ funksiyalari bilan parsing qilish.

- **parse_guess**(x, na = c("", "NA"), locale = default_locale())
- **parse_character**(x, na = c("", "NA"), locale = default_locale())
- **parse_datetime**(x, format = "", na = c("", "NA"), locale = default_locale()) va **parse_date()** hamda **parse_time()**
- **parse_double**(x, na = c("", "NA"), locale = default_locale())
- **parse_factor**(x, levels, ordered = FALSE, na = c("", "NA"), locale = default_locale())
- **parse_integer**(x, na = c("", "NA"), locale = default_locale())
- **parse_logical**(x, na = c("", "NA"), locale = default_locale())
- **parse_number**(x, na = c("", "NA"), locale = default_locale())

x\$A <- parse_number(x\$A)

Tibble - mukammallashgan data frame

tibble paketi jadval ma'lumotlarini saqlash uchun data frame vorisi bo'lgan tibble klassini taqdim qiladi. U quyidagilari bilan afzal:

- **Ko'rsatish** - Tibble chop qilinganda, undagi ma'lumot qisqa shaklda ekranga sig'adigan qilib chiqariladi.
- **Kichik to'plamga ajratish** - [har doim tibble qaytaradi, [[va \$ doimo vektor qaytaradi.
- **Qisman mos kelish mavjud emas** - to'plamga ajratishda ustunning to'liq nomini keltirishingiz lozim.

```
# A tibble: 234 x 6
  manufacturer <chr> model <chr> displ <dbl>
1 audi a4 1.8
2 audi a4 1.8
3 audi a4 2.0
4 audi a4 2.0
5 audi a4 2.8
6 audi a4 2.8
7 audi a4 3.1
8 audi a4 quattro 1.8
9 audi a4 quattro 1.8
10 audi a4 quattro 2.0
# ... with 224 more rows, and 3
# more variables: year <int>,
# cyl <int>, trans <chr>
```

tibble shaklida

```
156 1999 6 auto(l4)
157 1999 6 auto(l4)
158 2008 6 auto(l4)
159 2008 8 auto(s4)
160 1999 4 manual(m5)
161 1999 4 auto(l4)
162 2008 4 manual(m5)
163 2008 4 manual(m5)
164 2008 4 auto(l4)
165 2008 4 auto(l4)
166 1999 4 auto(l4)
# reached getOption("max.print")
# -- omitted 68 rows --
```

Ko'rsatilayotgan katta jadval

data frame shaklida

- Odatiy holni quyidagicha moslash mumkin:
options(tibble.print_max = n, tibble.print_min = m, tibble.width = Inf)
- Ma'lumotlarni ko'rish **View(x, title)** yoki **glimpse(x, width = NULL, ...)**
- Data framega o'tkazish **as.data.frame()** (ba'zi eski paketlar uchun zarur)

tibbleni 2 xil usulda yaratish

tibble(...)
Ustunlar bo'yicha
tibble(x = 1:3, y = c("a", "b", "c"))

Ikkalasi ham ushbu tibbleni yaratadi

tribble(...)
Satrlar bo'yicha
tribble(~x, ~y, 1, "a", 2, "b", 3, "c")

```
A tibble: 3 x 2
  x     y
<int> <dbl>
1     1  a
2     2  b
3     3  c
```

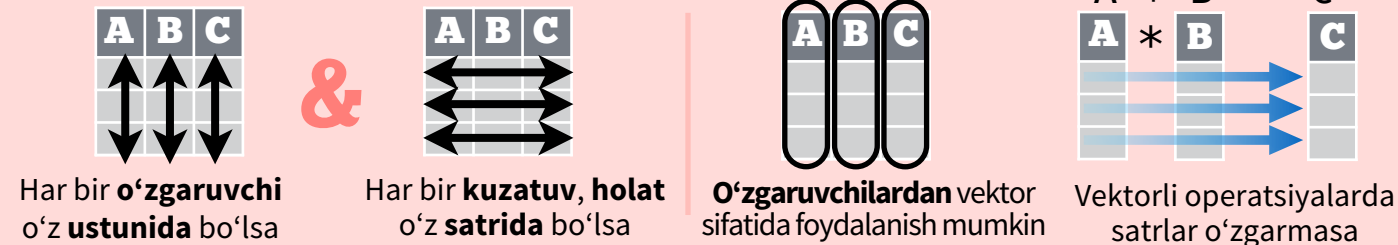
as_tibble(x, ...) data frameni tibblega aylantirish

enframe(x, name = "name", value = "value")
Vektorni "name" va "value" ustunli tibblega o'zgartirish.

is_tibble(x) x tibbleligini tekshirish.

tidyr yordamida ma'lumotni tozalash

Toza ma'lumot bu jadvali ma'lumotlarni tashkillashtirish yo'li. U paketlararo o'zgarma tuzilmani ta'minlaydi. Jadval toza bo'ladi qachonki:



Ma'lumotlar shaklini o'zgartirish - jadvalda ma'lumot keltirilishini o'zgartirish

gather() va **spread()** dan jadval ma'lumotlarini qayta taqsimlashda qo'llash mumkin. Bunda, key : value juftligidan foydalaniladi.

gather(data, key, value, ..., na.rm = FALSE, convert = FALSE, factor_key = FALSE)

Gather, ustun nomlarini **key** ustuniga, satr qiymatlarini **value** ustuniga ko'chiradi.

table4a

country	1999	2000
A	0.7K	2K
B	37K	80K
C	212K	213K

key value

gather(table4a, `1999`, `2000`, key = "year", value = "cases")

spread(data, key, value, fill = NA, convert = FALSE, drop = TRUE, sep = NULL)

Spread, **key** ustunining unikal qiymatlarini ustun nomlariga aylantiradi va **value** ustuni qiymatlarini yangi ustunga joylaydi.

table2

country	year	type	count
A	1999	cases	0.7K
A	1999	pop	19M
A	2000	cases	2K
A	2000	pop	20M
B	1999	cases	37K
B	1999	pop	172M
B	2000	cases	80K
B	2000	pop	174M
C	1999	cases	212K
C	1999	pop	1T
C	2000	cases	213K
C	2000	pop	1T

key value

spread(table2, type, count)

Yo'q qiymatlar ustida ishlash

drop_na(data, ...)

NAdan iborat satrlarni tashlab yuborish.

x

x1	x2
A	1
B	NA
C	NA
D	3
E	NA

drop_na(x, x2)

fill(data, ..., .direction = c("down", "up"))

NAlarni eng yaqin qiymatga o'zgartirish

x

x1	x2
A	1
B	1
C	1
D	3
E	3

fill(x, x2)

replace_na(data, replace = list(), ...)

Ustunga ko'ra almashtirish.

x

x1	x2
A	1
B	2
C	2
D	3
E	2

replace_na(x, list(x2 = 2), x2)

Jadvallarni kengaytirish - qiymatlar to'plami bilan jadval yaratish

complete(data, ..., fill = list())

...da keltirilgan o'zgaruvchilarning mavjud bo'lmagan qiymatlar to'plamini asosiy ma'lumotlarga qo'shish

complete(mtcars, cyl, gear, carb)

expand(data, ...)

...da keltirilgan o'zgaruvchilarning mumkin bo'lgan qiymatlaridan iborat yangi tibble yaratish

expand(mtcars, cyl, gear, carb)

Ajratish va birlashtirish

Ushbu funksiyalardan yacheykalarni alohida qiymatlarga ajratish yoki birlashtirish uchun foydalaning.

separate(data, col, into, sep = "[^[:alnum:]]+", remove = TRUE, convert = FALSE, extra = "warn", fill = "warn", ...)

Ustundagi har bir yacheykani ajratib bir nechta ustunlar hosil qilish.

table3

country	year	rate
A	1999	0.7K/19M
A	2000	2K/20M
B	1999	37K/172M
B	2000	80K/174M
C	1999	212K/1T
C	2000	213K/1T

→

country	year	cases	pop
A	1999	0.7K	19M
A	2000	2K	20M
B	1999	37K	172
B	2000	80K	174
C	1999	212K	1T
C	2000	213K	1T

separate(table3, rate, into = c("cases", "pop"))

separate_rows(data, ..., sep = "[^[:alnum:]]+", convert = FALSE)

Ustundagi har bir yacheykani ajratib, bir nechta qator yaratish. Shuningdek, **separate_rows()**.

table3

country	year	rate
A	1999	0.7K/19M
A	2000	2K/20M
B	1999	37K/172M
B	2000	80K/174M
C	1999	212K/1T
C	2000	213K/1T

→

country	year	rate
A	1999	0.7K
A	1999	19M
A	2000	2K
A	2000	20M
B	1999	37K
B	1999	172M
B	2000	80K
B	2000	174M
C	1999	212K
C	1999	1T
C	2000	213K
C	2000	1T

separate_rows(table3, rate)

unite(data, col, ..., sep = "_", remove = TRUE)

Bir nechta ustunlarni yagona ustunga birlashtirish.

table5

country	century	year
Afghan	19	99
Afghan	20	0
Brazil	19	99
Brazil	20	0
China	19	99
China	20	0

→

country	year
Afghan	1999
Afghan	2000
Brazil	1999
Brazil	2000
China	1999
China	2000

unite(table5, century, year, col = "year", sep = "")